USAF ACADEMY
DEPARTMENT OF ELECTRICAL AND COMPUTER ENGINEERING
LESSON 16 LAB - FREQUENCY HOPPING
Lab Packet
Spring 2027

Name: _____ **Section:** _____ **Date:** _____

Learning Objective 6.5: I can implement FHSS transmission and reception, including hop-sequence synchronization.

This packet is the L16 Lab, in three parts, and you turn it in once, complete, before Lesson 17. Do **Part 1** on your own before the lab period: run the link flowgraph and make the predictions you will check at the bench. **Part 2** is your own measurements, recorded in class: Questions 3 to 5 on your own laptop, and Question 6 at a PlutoSDR station with your team. Do **Part 3** on your own after the lab: compare the measurements with the predictions and answer the analysis questions.

Grading. You submit the packet once, we grade it against Objective 6.5 on the course's 1 / 0 scale, and you may revise it under course policy. Paste screenshots where a question asks for one.

Transmitting. In class we transmit only into a cable, into a shielded enclosure, or at very low power in an ISM band, and only when the lesson says to. Questions 1 to 5 run on your laptop and transmit nothing. Question 6 transmits from a PlutoSDR's TX port, through a cable with an inline 30 dB SMA attenuator, to the same Pluto's RX port, at 915 MHz. Connect the cable and attenuator before you run the flowgraph, and never connect an antenna in this lab. Never set TX attenuation below 30 dB or change the slider's range; the RX input is rated to only a few dBm. We have run the PlutoSDR flowgraph only in simulation, so Question 6 is its first run on hardware.

The flowgraphs are on the L16 lesson page, `L16-fhss-tx.grc` and `L16-fhss-link.grc`, and on the L16 Lab page, `L16-fhss-pluto.grc`. Every number below assumes their settings: Lesson 8's FSK at 1000 baud, sampled at 128 kHz; one packet per hop, made of `guard` guard bytes, the 12-byte access code and header, the 14-byte payload `USAFA ECE 448` with its newline, and `guard` more guard bytes; N channels 10 kHz apart; and key 448, whose sync channel for $N = 8$ is channel 7. With one channel of N jammed, the fraction of packets delivered is about

$$P = 1 - \frac{1}{N}.$$

Part 1: Pre-Lab (individual, before the lab period)

1. **Run the link.** Open `L16-fhss-link.grc` in GRC and run it with every slider at its default.

   (a) Once the Receiver label reads "locked," wait 30 s and record the Receiver and Packets labels.

   (b) Paste a screenshot of the running flowgraph's window.

2. Predictions. You will check each of these at the bench.

(a) Set `guard` to 4 bytes. Find the bits in one hop, the time on one hop, the hop rate, the time for one cycle of 8 channels, and the samples in one hop at 128 kHz.

	Bits per hop	Hop time	Hop rate	Cycle	Samples
`guard` = 4					

(b) With `guard` = 4, about how early and how late can the receiver change channel and still receive every packet?

(c) For the hopper you build in Question 3, choose your own order of the 8 channels, using each once, and write the hop frequencies in that order. With `dwell` = 32,000 samples at 128 kHz, find the time on one hop and the time for one cycle.

(d) A jammer strong enough to take every hop on its channel sits on one channel that is not the sync channel. Predict the fraction of packets delivered with 2 and with 4 channels. Then, for a weaker jammer that lets a fraction p through when the link stays on its channel, write the fraction delivered when the link hops over 8 channels.

Channels	2	4	8, weak jammer
Delivered			

Part 2: Measurements (in class)

3. **Build a hopper and a de-hopper.** Start GRC with an empty canvas. Do not open `L16-fhss-tx.grc`: build the hopper and the de-hopper yourself as the lab page lists, with your order from Question 2(c).

    (a) Paste a screenshot of your running flowgraph's window with no Delay block. Mark where the jammer appears on each waterfall.

    (b) Paste a screenshot with the Delay block set to `dwell`. Explain what the De-hopped waterfall shows.

    (c) Describe the De-hopped waterfall with the delay at 2000 samples.

4. Jamming. Follow the lab page's steps for `L16-fhss-link.grc`. Record the Packets label after 30 s at each setting, as correct of sent, and as a percent.

(a) Fill in the table.

`nch`	8	1	8	2	4
Jammer	2, ch 4	0.7	0.7, ch 4	2, ch 1	2, ch 1
Delivered					

(b) Use your fixed-channel result at 0.7 as p in Question 2(d), and compare the prediction with the hopping result.

(c) Explain why the delivered fraction at 2 and 4 channels matches Question 2(d).

(d) Record the Receiver and Packets labels with the jammer at 3 on channel 7 turned on after the receiver locked, and again turned on before it locked.

5. Sync. Follow the lab page's steps.

(a) Record the Packets label with the hop offset at $+1$ and at -1.

(b) With `guard` = 4, fill in the table, and compare it with Question 2(b).

Timing (ms)	-24	$+24$	-30	$+30$	-36	$+36$
Delivered						

(c) Record the three times from moving the path delay to the next received line.

6. **Frequency hopping on a PlutoSDR.** At a PlutoSDR station, with your team, follow the lab page's steps: connect the Pluto's TX port to its RX port through the 30 dB attenuator, check the Pluto with `iio_info -s`, and run `L16-fhss-pluto.grc`. When you finish, stop the flowgraph and unplug the Pluto before you disconnect the cable.

   (a) Once `USAFA ECE 448` decodes, paste a screenshot that shows both waterfalls and the Receiver, Packets, and Received labels.

   (b) Record the Packets label with the jammer at 2 on channel 4.

   (c) Record the Receiver and Packets labels at each TX attenuation, and explain what limits the link as the attenuation rises.

TX attenuation (dB)	60	70	80	89
Receiver				
Delivered				

Part 3: Writeup (individual, after the lab)

7. Analysis.

(a) Compare your jamming table with Question 2(d). Use your De-hopped waterfall from Question 3(a) to explain why a jammer stronger than the signal takes only the hops on its own channel, and why raising its power further does not take more.

(b) Compare your timing windows with 2 and 4 guard bytes. Find the share of each hop that carries the payload in each case, and explain the tradeoff.

(c) From Question 4(d), explain why one jammed channel can keep a receiver from ever locking although it costs a locked receiver only 1 hop in 8, and propose a change to Hop Sync or to the packet that removes that weakness.

Documentation: _____